ARMAAN CHOUDHARY

SOFTWARE ENGINEER & SYSTEMS ARCHITECT

Delhi NCR, India • **Portfolio: armaan.works** • **GitHub: github.com/armaan-choudhary** • **LinkedIn: armaan-choudhary**

PROFESSIONAL SUMMARY

Software Engineer based in Delhi NCR specializing in full-stack architecture, high-performance UI systems, and secure AI integrations. Demonstrated track record building multi-agent RAG platforms, quantized Edge AI vision models, and production web systems. Experienced in leading team delivery, implementing strict application security (RBAC, constant-time hashing, input sanitization), and optimizing end-to-end latency with automated testing pipelines.

TECHNICAL SKILLS

Languages:	JavaScript (ES6+), TypeScript, Python, PHP 8.2+ (OOP), SQL, C++
Frontend Architecture:	Next.js 16 (App Router), React 19, Tailwind CSS v4, Framer Motion, GSAP 3, Swup.js
Backend & Systems:	FastAPI, Node.js, RESTful APIs, PostgreSQL & pgvector, MySQL, Docker Compose, Supabase
AI & Retrieval (RAG):	LangGraph (Multi-Agent), PyTorch, TensorRT, INT8 Quantization, Google Gemini 2.5, Sentence Transformers, FAISS, BM25, Llama.cpp
DevOps & Testing:	Pytest (Automated Testing), Vite, Git, Linux, Vercel, CI/CD Pipeline Automation

ENGINEERING EXPERIENCE

Client Engineering Projects — *Full-Stack Engineer & Architect* **2026**

- Delivered client production applications across **FortuneWander** and **QueenTrends Theme** with 0-page-reload SPA navigation.
- Architected custom PHP asset loaders parsing dynamic Vite build manifests, eliminating render-blocking styles and cutting image payloads by 80%.
- Implemented strict whitelist-based HTML sanitizers (`wp_kses`) and transient-based IP rate limiting for secure Google Gemini LLM API integration.
- Achieved **95+ Lighthouse performance scores**, sub-second Largest Contentful Paint (LCP), and full mobile responsiveness.

FEATURED PROJECTS

CampusOne (msInnovateHacks) — *Project Lead, Frontend & Security* (Team of 3) **2026**

- Led team of 3 to engineer an enterprise multi-agent university front door, backed by a **54-test automated suite** (`pytest`).
- Built the dual-role **Next.js 16 (App Router) / React 19** frontend featuring an obsidian dark student workspace with docked citation inspection drawers (2x2 metadata grid with page-level PDF provenance) and live administrative triage console (`/admin`).
- Architected application security: constant-time PBKDF2 password hashing (100k rounds), signed HMAC JWTs, HTTP 403 RBAC barriers, and horizontal access isolation enforcing server-derived thread ownership.
- Co-designed the **LangGraph StateGraph** router with margin-guarded execution (`≥0.75` direct routing vs `<0.75` clarification loops) across 4 isolated PostgreSQL + `pgvector` knowledge stores with Maximal Marginal Relevance (MMR).

PrismNet (KJP-66) — *AI & Systems Engineer* **2026**

- Compressed an RT-DETR Vision Transformer, slashing inference latency from **133ms to 21ms (6.3× speedup)** and achieving **47.5 FPS** on an entry-level mobile RTX 2050 GPU.
- Applied PyTorch L1 unstructured pruning (30% sparsity) with INT8 quantization, cutting model size by **49.7% (130MB to 66MB)** while maintaining 0.528 mAP@0.5:0.95; compiled natively into a C++ **NVIDIA TensorRT** engine to eliminate CPU-bound NMS bottlenecks.

FortuneWander — *Full-Stack Developer & Architect* **2026**

- Engineered a custom WordPress travel portal decoupling the frontend with **Swup.js**, **Lenis Scroll**, and **Vite** for seamless 0-page-reload SPA navigation.
- Built an object-oriented PHP AI Travel Consultant plugin integrating **Google Gemini 2.5** with search grounding and automated day-by-day HTML itineraries.
- Engineered secure custom REST API endpoints with WordPress Nonce authentication, preventing API key exposure and automating sanitization.

FinChat — *AI Engineer & Architect* **2026**

- Built an asynchronous 100% offline local RAG system for financial analysts utilizing **FastAPI**, **Python**, **FAISS**, and **Llama.cpp**.
- Engineered a hybrid retrieval pipeline combining dense vector embeddings with **BM25Okapi** keyword ranking, fed into a cross-encoder reranker that eliminated **>90% of irrelevant chunks** and preserved exact financial figures.

EDUCATION

Bennett University — *Bachelor of Technology (Computer Science & Engineering)* **2024 – 2028**

- High academic standing; demonstrated mastery in Data Structures, Algorithms, System Architecture, and Software Engineering.

The Millennium School — *CBSE High School Examination* **2020 – 2022**

- High distinction with A+ grades in Mathematics, Computer Science, and Physics.